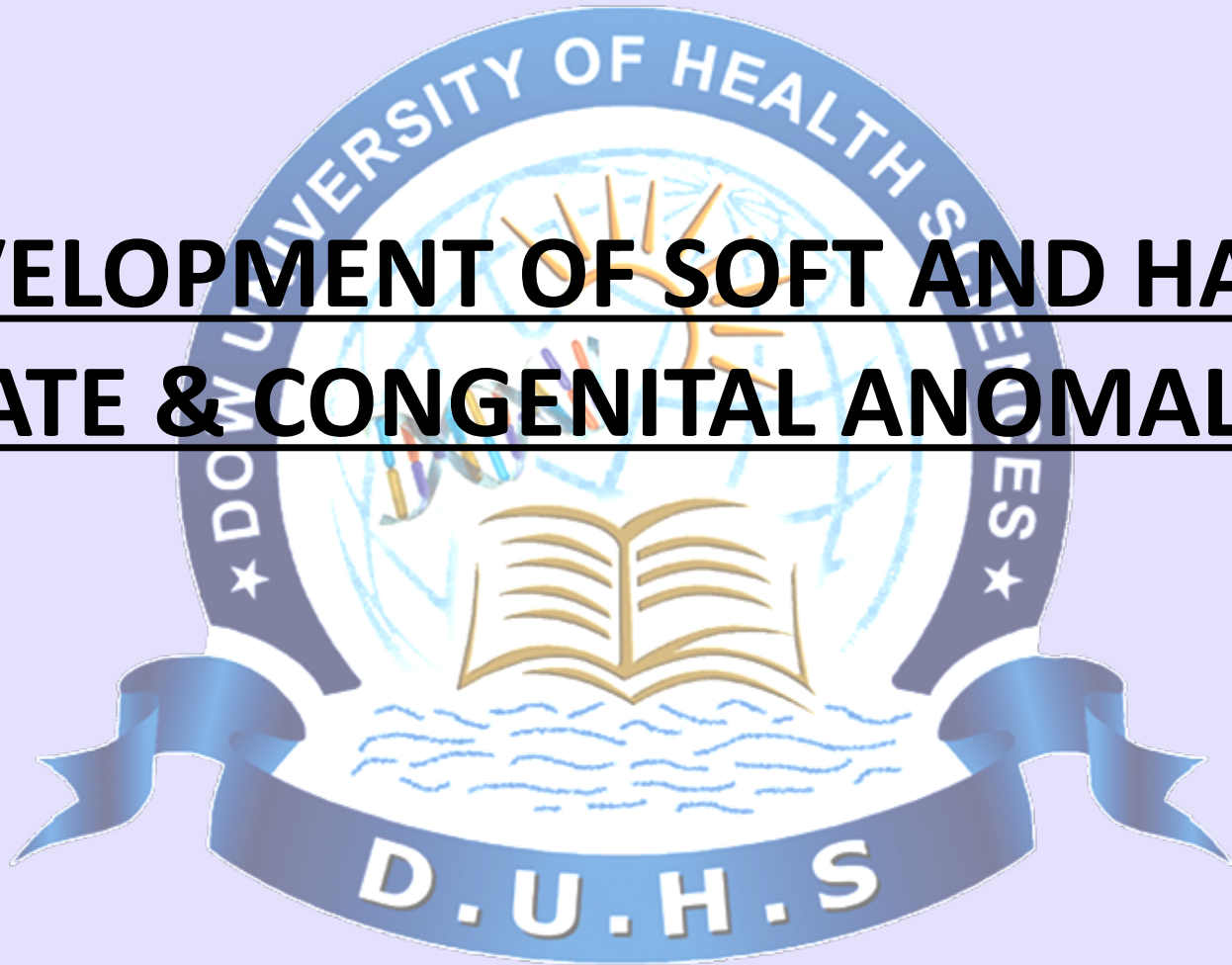
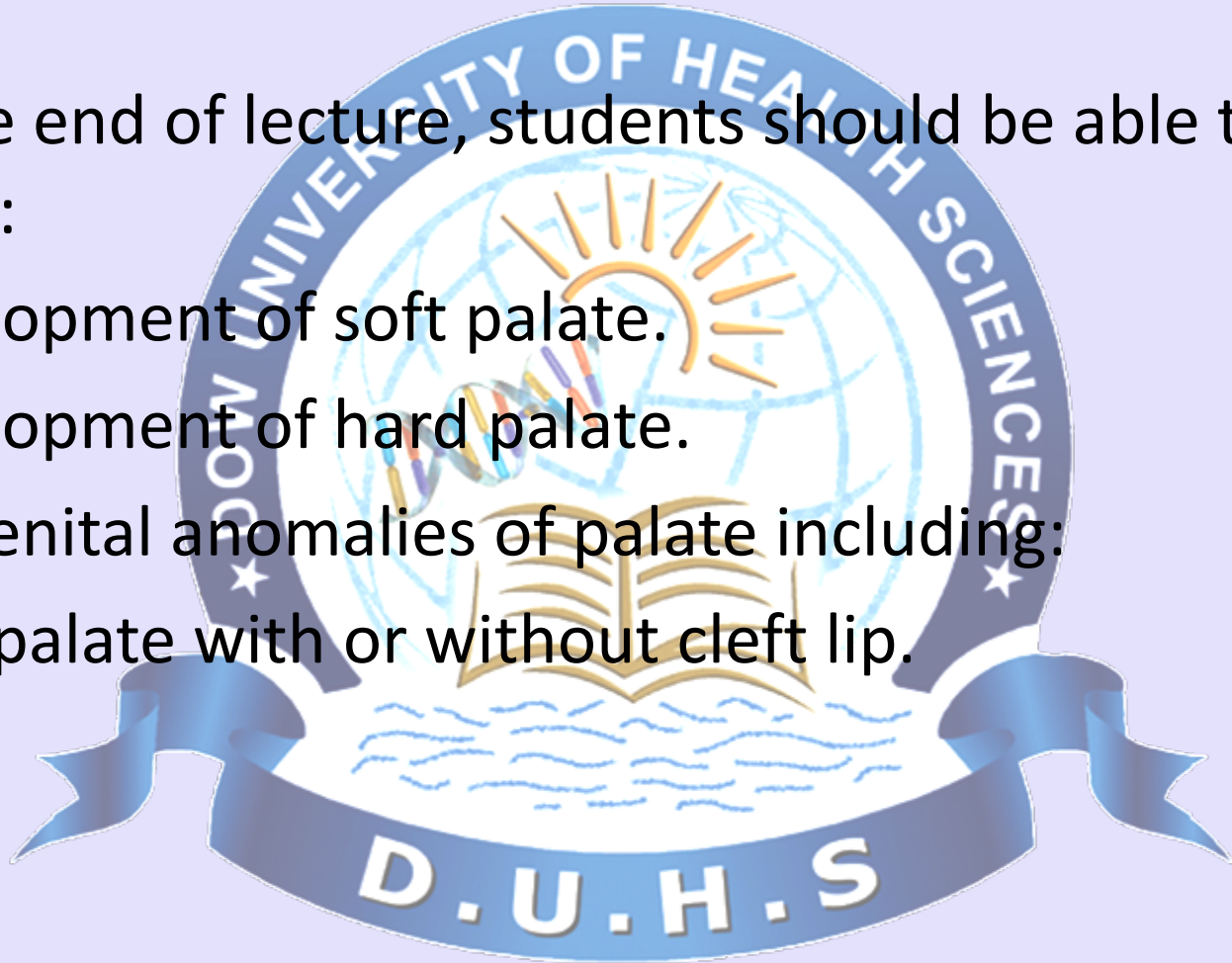


DEVELOPMENT OF SOFT AND HARD PALATE & CONGENITAL ANOMALIES.



Learning Objectives.

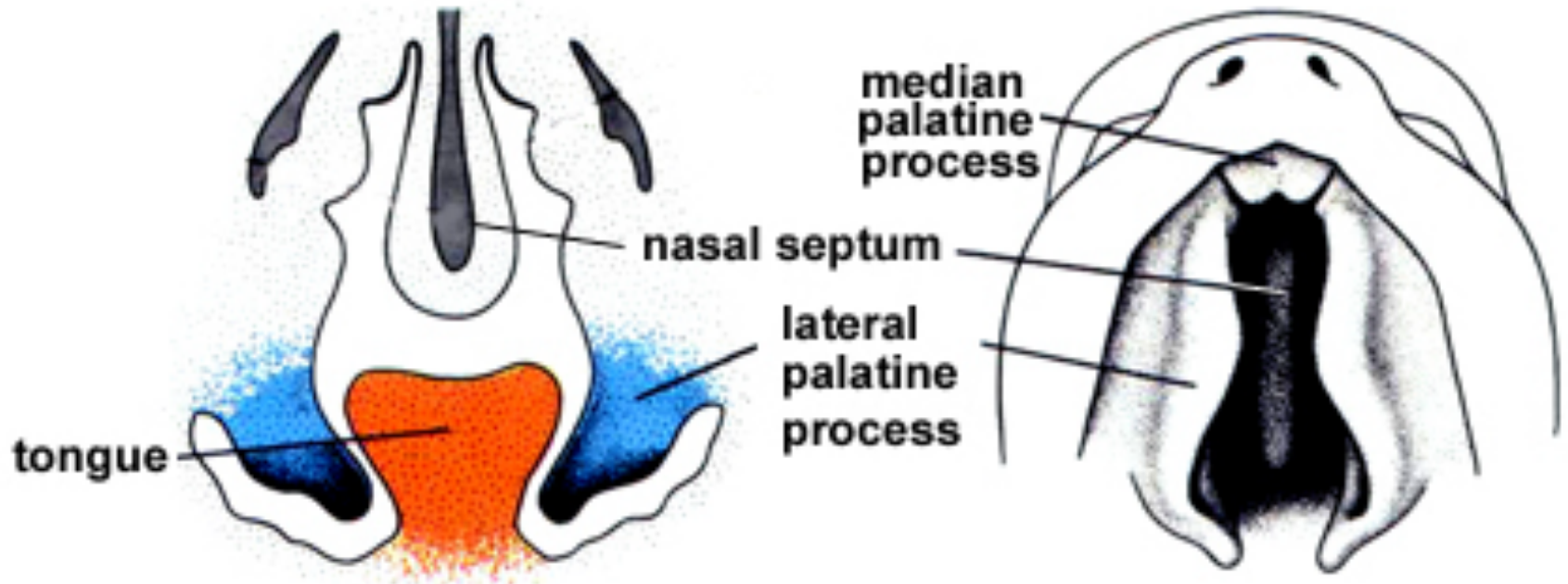
- At the end of lecture, students should be able to know:
- Development of soft palate.
- Development of hard palate.
- Congenital anomalies of palate including:
- Cleft palate with or without cleft lip.



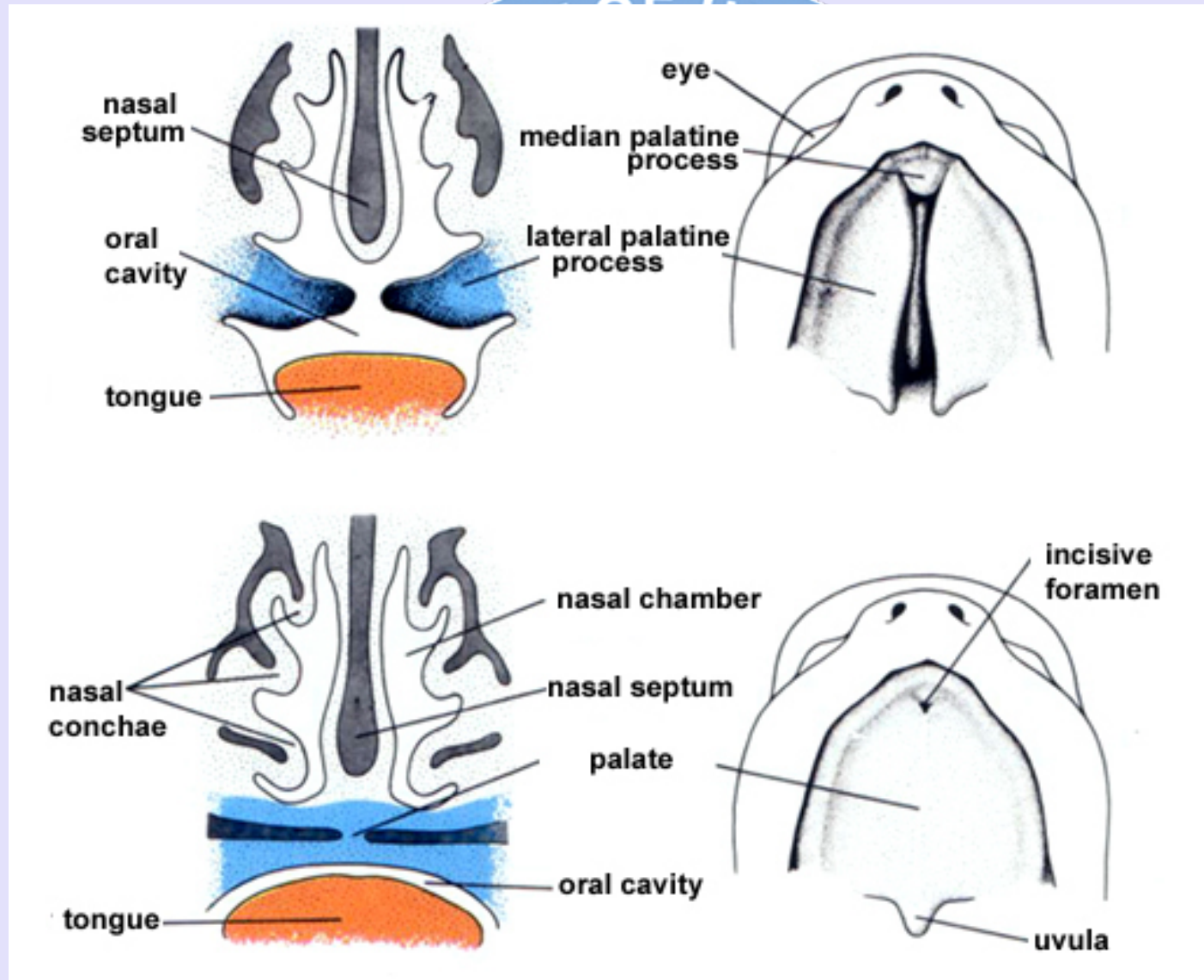
Development of Palate (5th - 12th week).

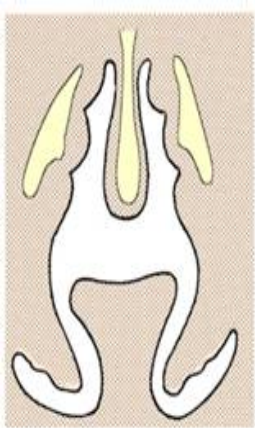
Primordia:

- Median palatine process.
- Lateral palatine processes.

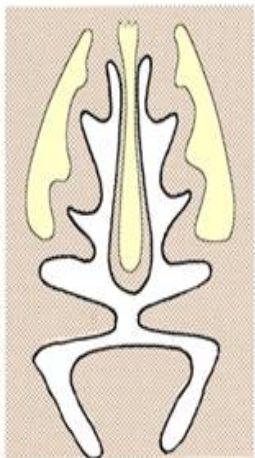
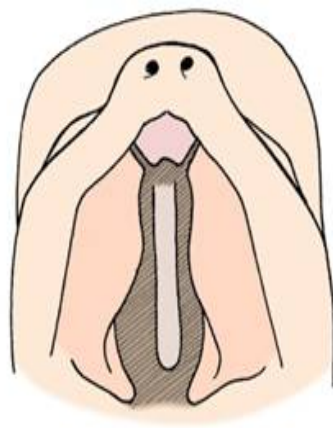


Process: lateral and median palatine processes fuse → formation of palate.

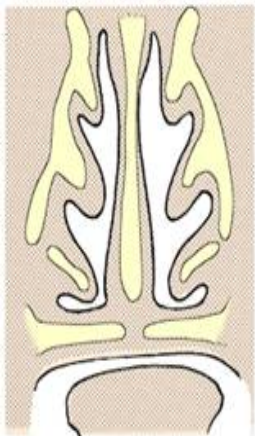
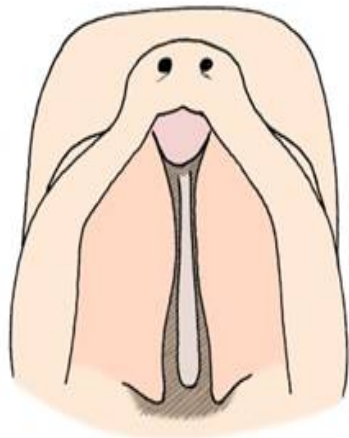




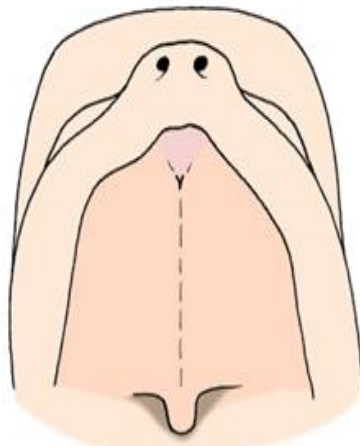
A



B



C

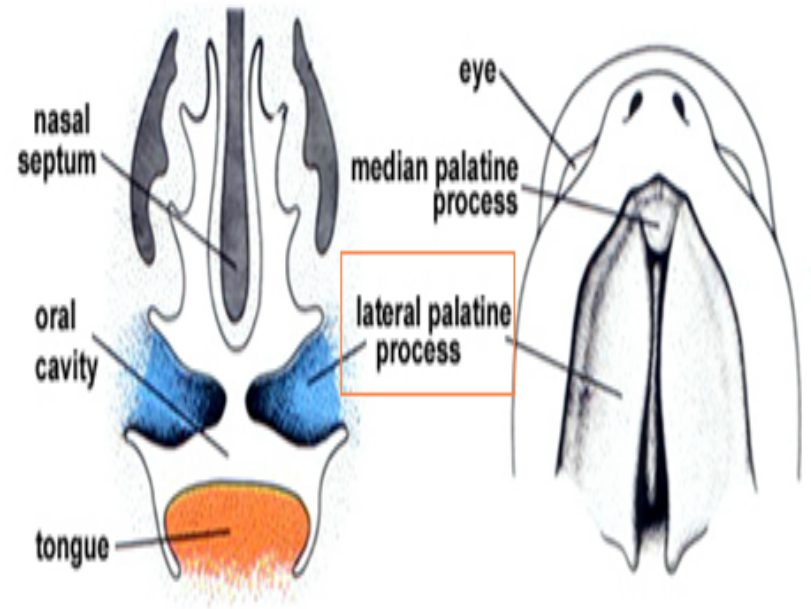


Median palatine process: or primary palate:

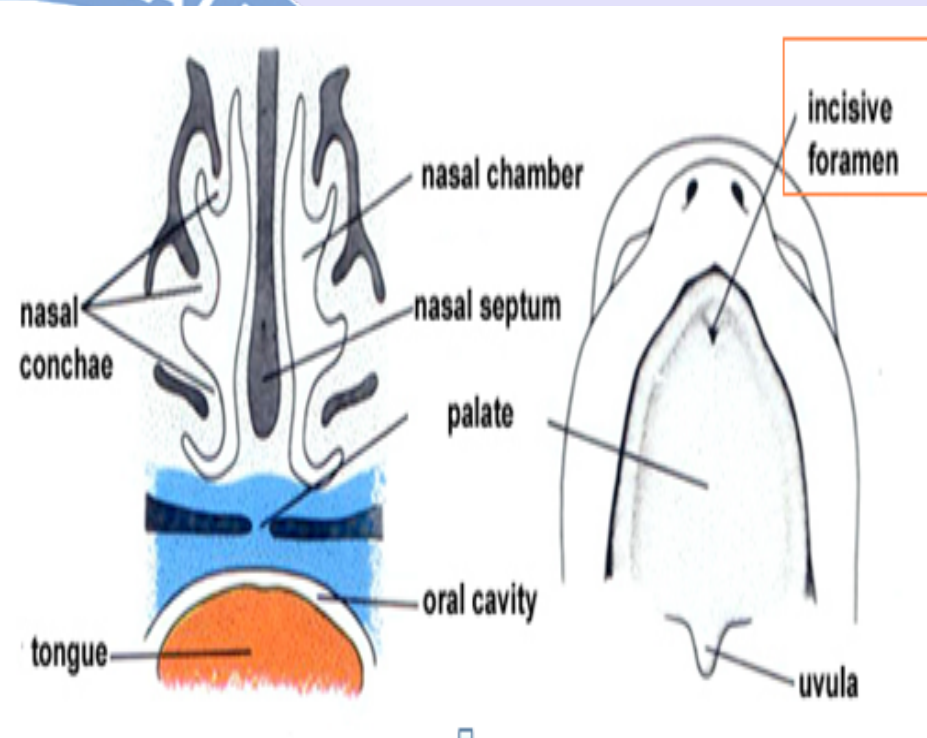
- Develops by 6th week.
- Mesenchymal.
- Proliferation of inner portion of intermaxillary segment.
- To give rise to small portion of palate.

lateral palatine process:

- Develops by 8-9th week.
- Mesenchymal.
- proliferation of inner portion of maxillary prominence.
- It fuses with each other, then fuses with the triangular primary palate to form secondary palate.



- **Incisive foramen:** the midline landmark between the primary and secondary palates
- **Hard palate:** ossification of anterior portion of secondary palate
- **Soft palate:** posterior portion of secondary palate

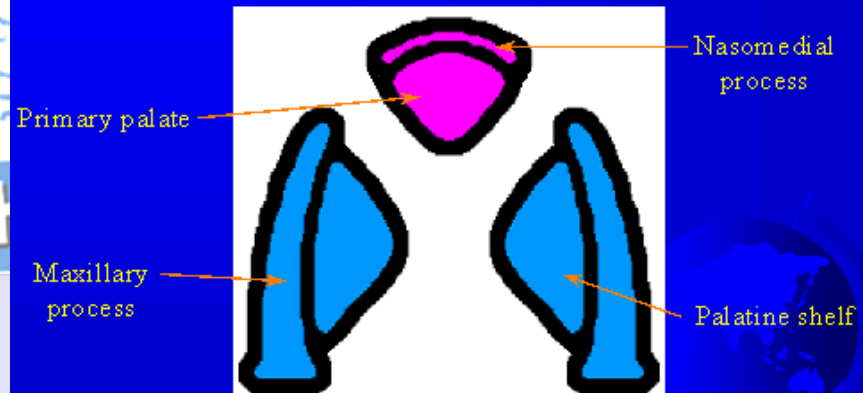


6-7th Week (GA).

- The floor of the oral cavity at this stage is known as primary palate.
- The medial walls of maxillary swellings begin to produce a pair of thin medial extensions “palatal shelves”, which grow on either side of the tongue.



Intermaxillary Segment

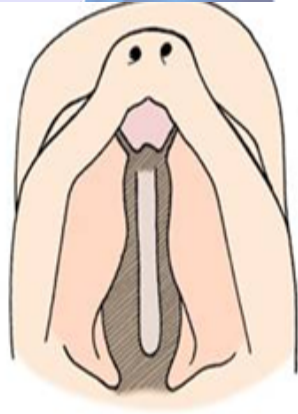


8th Week.

- The tongue moves downward and the palatine shelves rapidly rotate upward toward the midline, growing horizontally.



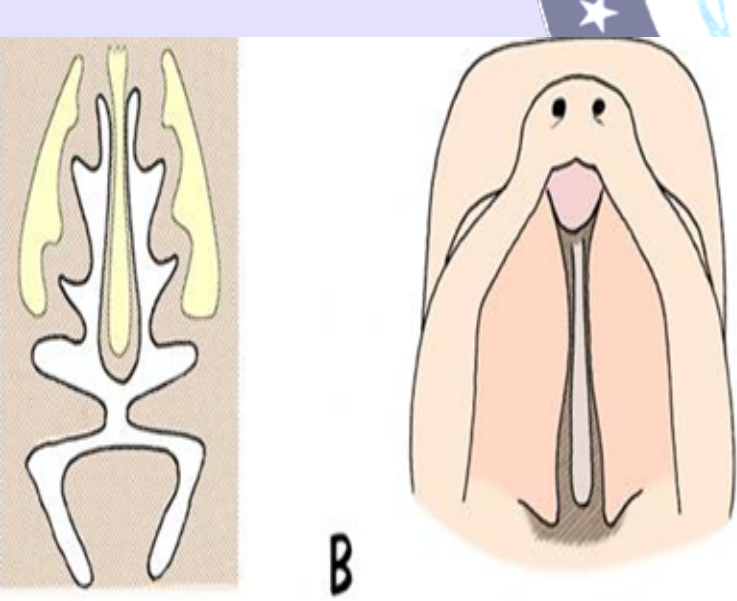
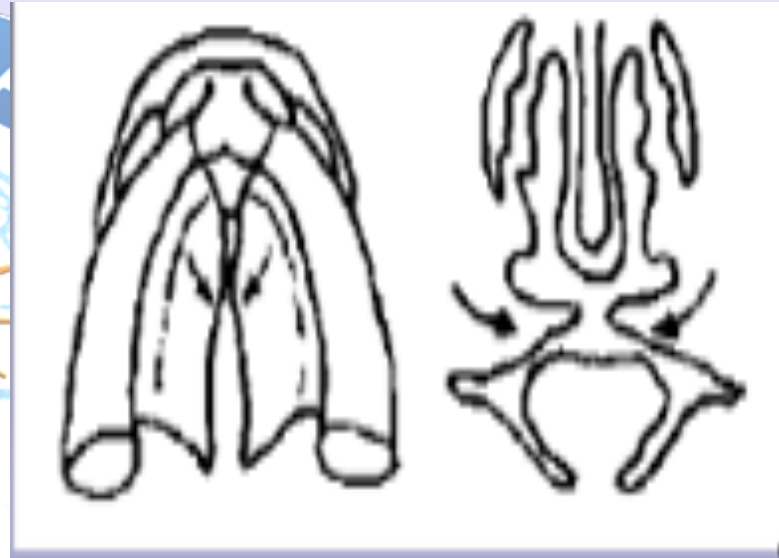
A



J.H.S

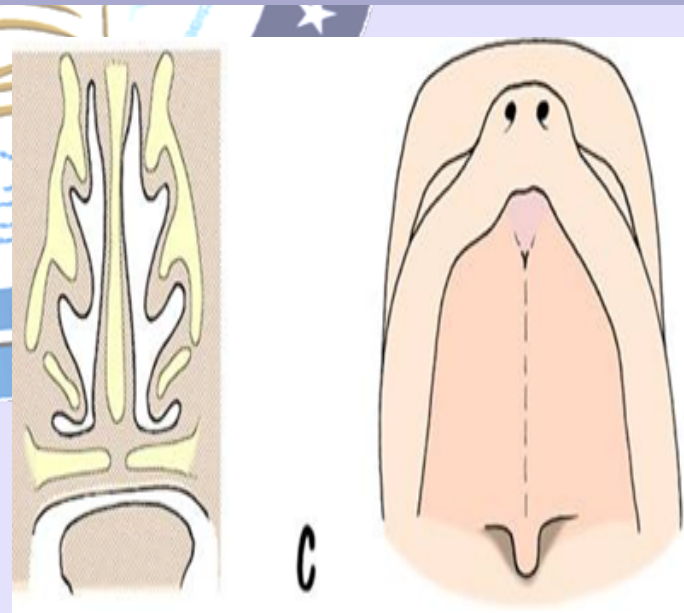
9th Week.

- The palatine shelves begin to fuse ventrodorsally with each other, the primary palate and the inferior nasal septum.



10th Week.

- The ventral secondary palate becomes the bony hard palate through mesenchymal condensations (endochondral ossification).
- The dorsal secondary palate becomes the soft palate through myogenic mesenchymal condensation.









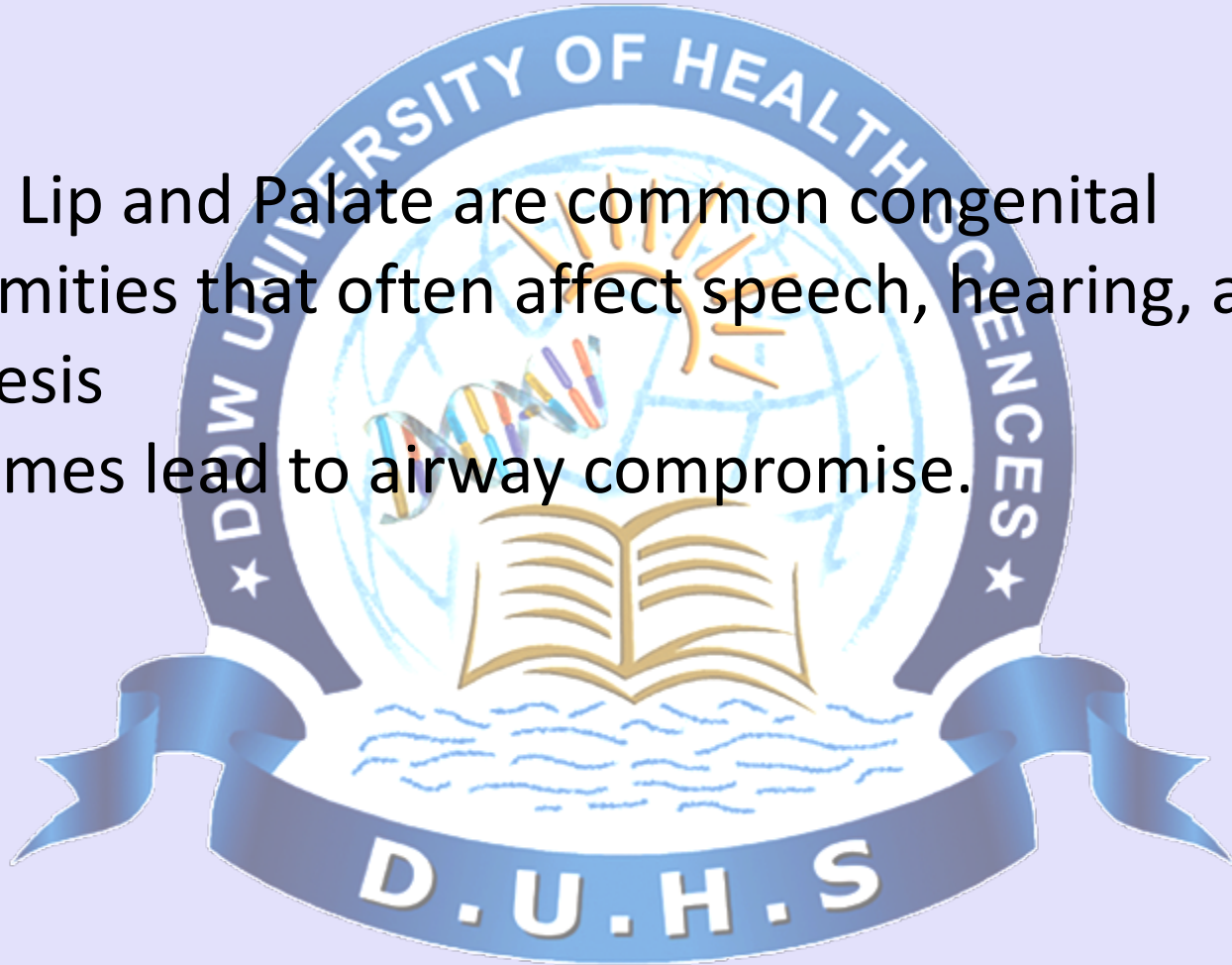


Congenital anomalies.



Cleft Palate.

- Cleft Lip and Palate are common congenital deformities that often affect speech, hearing, and cosmesis
- At times lead to airway compromise.



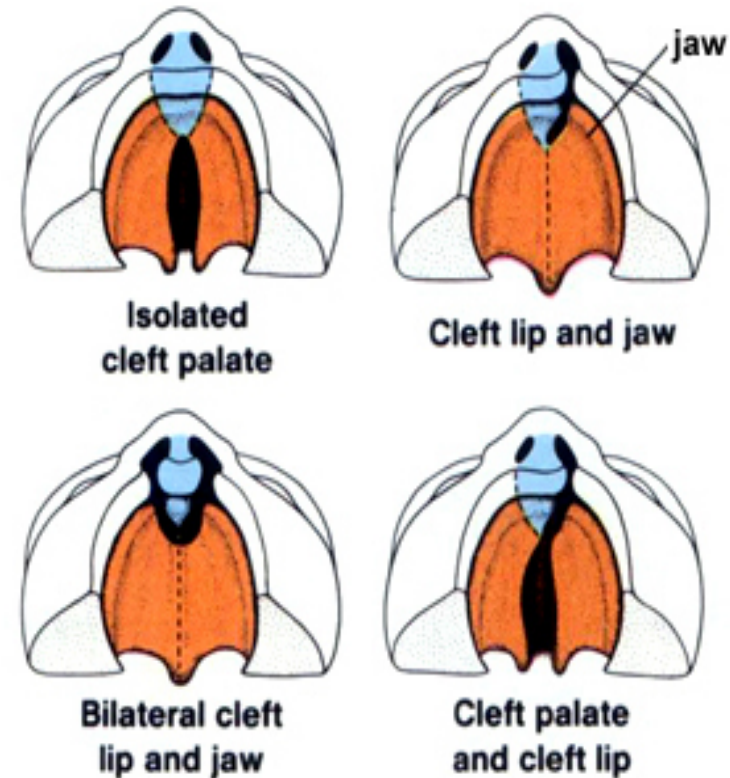
Cleft palate.

- Failure of lateral palatine process to fuse with each other or with median palatine process.

- Unilateral or bilateral, complete or incomplete, with or without cleft lip.

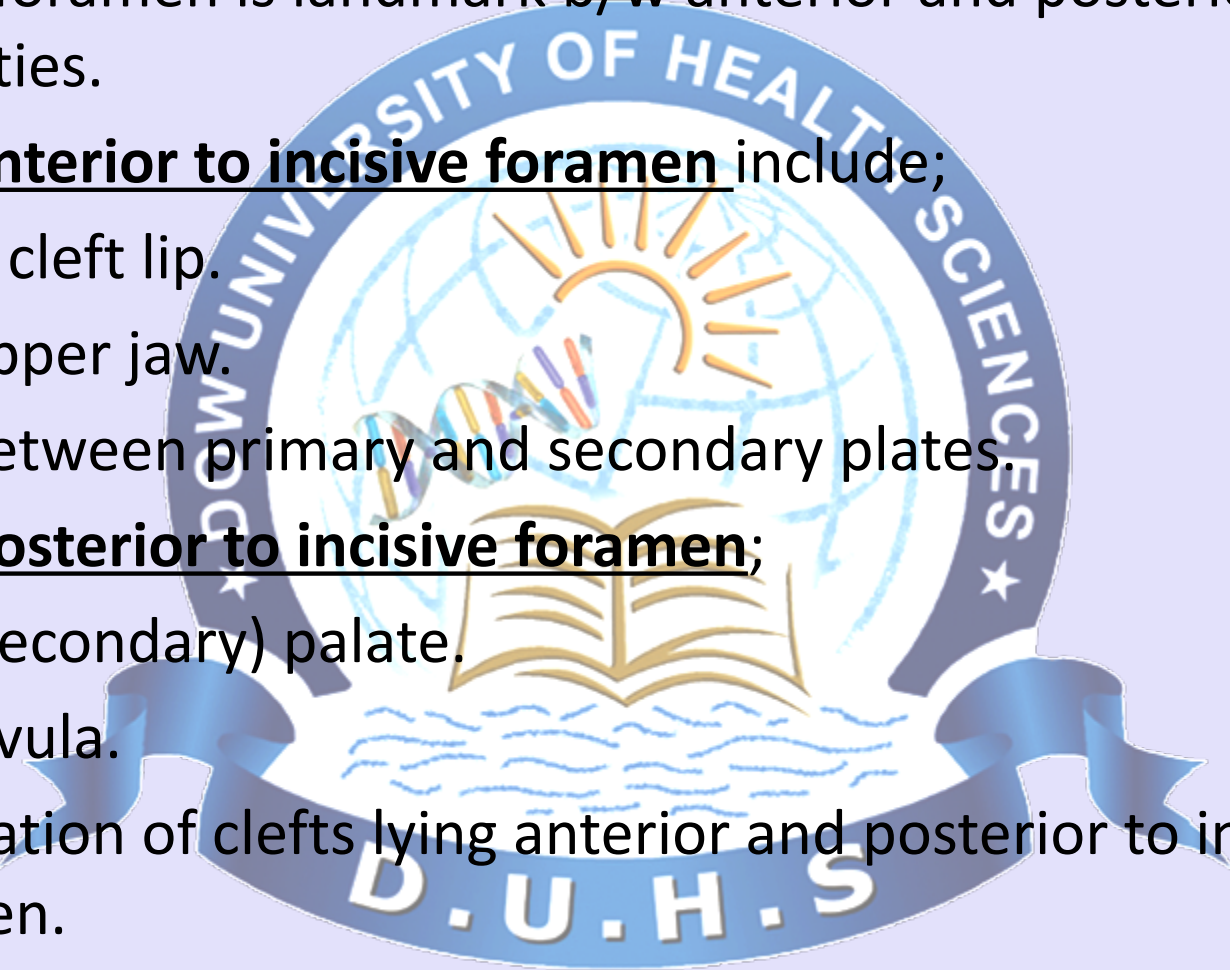
It can be:

- a. cleft of the primary palate
- b. cleft of secondary palate
- c. cleft of the primary and secondary palates.



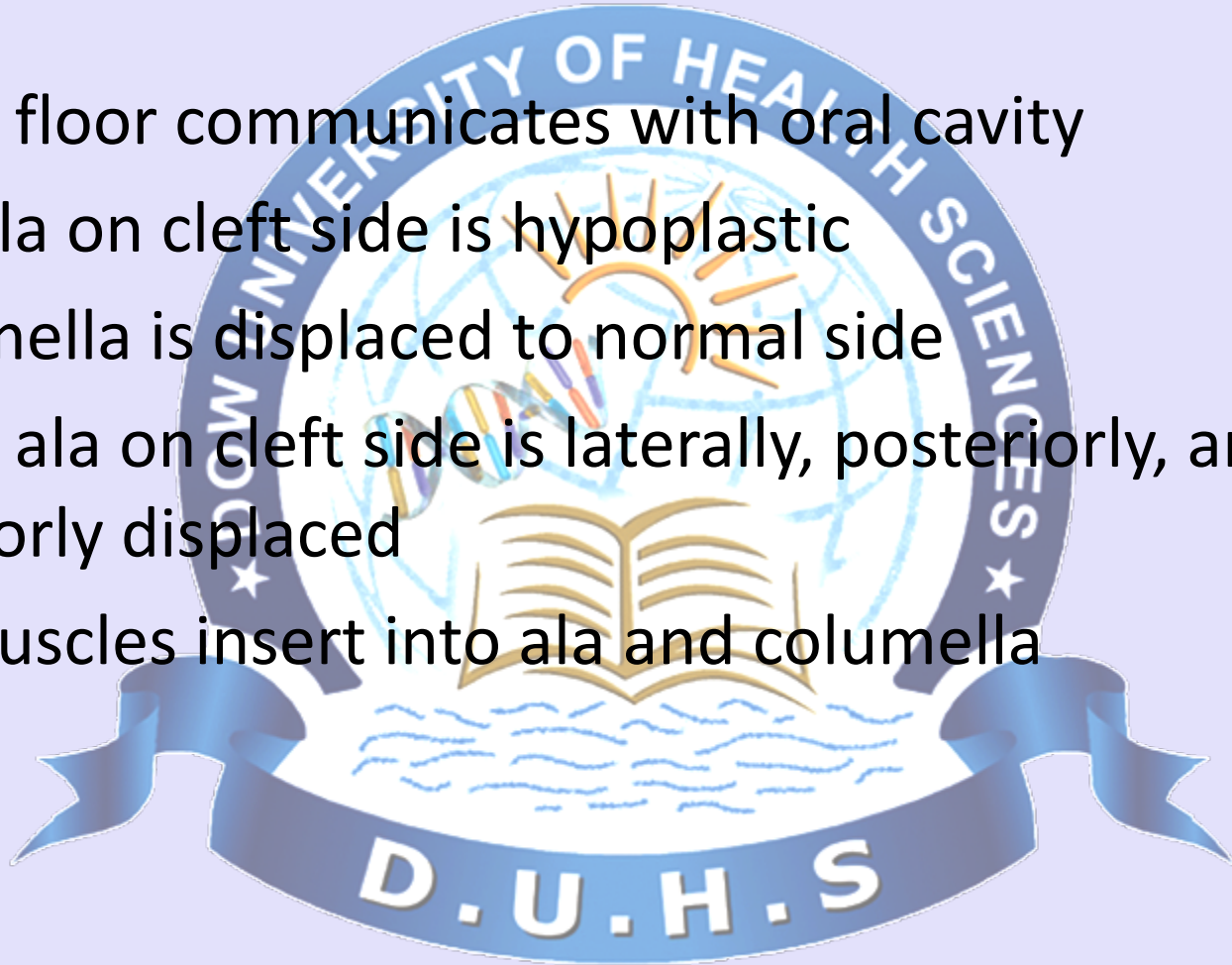
Cleft lip and palate.

- Incisive foramen is landmark b/w anterior and posterior cleft deformities.
1. Those **anterior to incisive foramen** include;
 - Lateral cleft lip.
 - Cleft upper jaw.
 - Cleft between primary and secondary plates.
 2. Those **posterior to incisive foramen**;
 - Cleft (secondary) palate.
 - Cleft Uvula.
 3. Combination of clefts lying anterior and posterior to incisive foramen.



Unilateral Cleft Lip.

- Nasal floor communicates with oral cavity
- Maxilla on cleft side is hypoplastic
- Columella is displaced to normal side
- Nasal ala on cleft side is laterally, posteriorly, and inferiorly displaced
- Lip muscles insert into ala and columella













UNILATERAL INCOMPLETE



UNILATERAL COMPLETE



BILATERAL COMPLETE



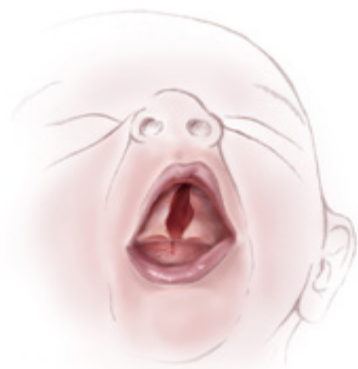
INCOMPLETE CLEFT PALATE



**UNILATERAL COMPLETE LIP AND
PALATE**



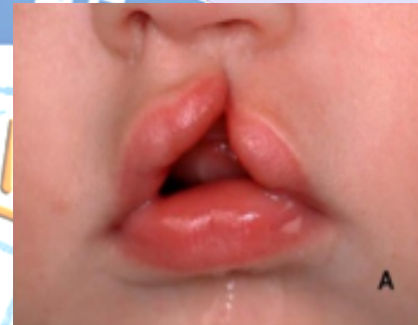
BILATERAL COMPLETE



Baby with cleft palate



Cleft palate



A



B



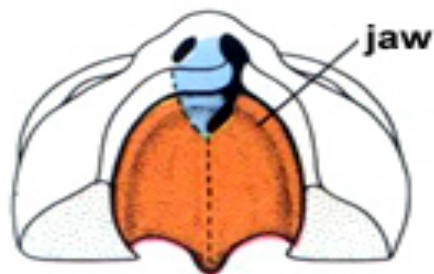
C



D



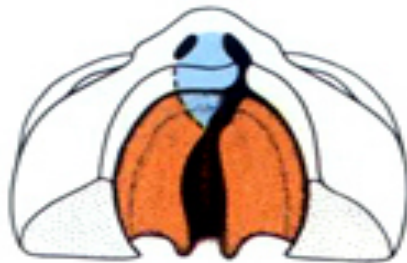
Isolated cleft palate



Cleft lip and jaw



Bilateral cleft lip and jaw

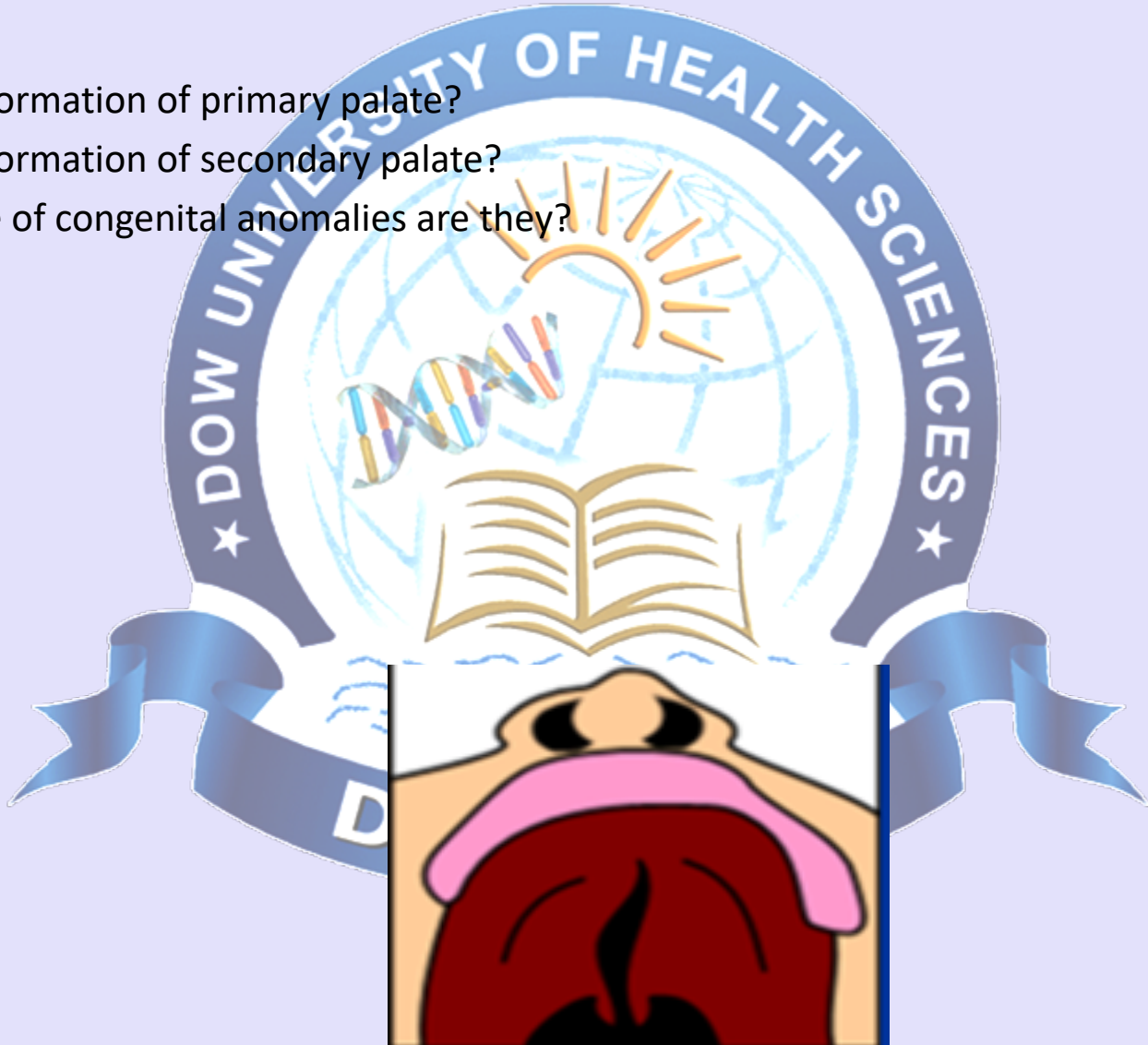


Cleft palate and cleft lip

Self Assessment.

1. Describe formation of primary palate?
2. Describe formation of secondary palate?
3. What type of congenital anomalies are they?

A.



Self Assessment.

- B.



- C.





Thank You